

L26 Power Series and Analytic Functions

Section 8.2 Series Review

Here a_n denotes the entire term, not just the coefficient.

Test for Divergence:

If $\lim_{n \rightarrow \infty} a_n$ DNE or if $\lim_{n \rightarrow \infty} a_n \neq 0$ then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

P-series:

The p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges for $p > 1$ and diverges for $p \leq 1$.

Direct Comparison test:

Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

1. If $\sum b_n$ is convergent and $a_n \leq b_n$ for all n , then $\sum a_n$ is also convergent
2. If $\sum b_n$ is divergent and $a_n \geq b_n$ for all n , then $\sum a_n$ is also divergent

Limit Comparison test:

Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$ where c is a finite, positive number, then either both series converge or both diverge.

Alternating Series test:

For $b_n > 0$ with $\sum_{n=1}^{\infty} (-1)^n b_n$, if

(1) $b_{n+1} \leq b_n$ for all n

(2) $\lim_{n \rightarrow \infty} b_n = 0$

then the series is convergent.

Ratio test:

For $\sum_{n=1}^{\infty} a_n$, if $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \begin{cases} < 1 & \text{absolutely convergent} \\ = 1 & \text{the test is inconclusive} \\ > 1 & \text{the series diverges} \end{cases}$

Root test:

For $\sum_{n=1}^{\infty} a_n$, if $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} \begin{cases} < 1 & \text{absolutely convergent} \\ = 1 & \text{the test is inconclusive} \\ > 1 & \text{the series diverges} \end{cases}$

Differential equations that we have previously studied had solutions in terms of elementary functions such as polynomials, logs, exponentials, sines, and cosines. But many important equations have solutions that cannot be similarly expressed.

Taylor polynomial approximations suggest a possibility: suppose the differential equation along with its initial conditions permits computation of every possible derivative $y^{(n)}$ at the point of expansion x_0 . Are there conditions that guarantee that the Taylor polynomial converges to a solution?

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Then might we find a solution to the DE by representing it as a Taylor series expansion.

But first we must review the properties of a certain type of Taylor polynomial called the **power series**.

A power series about the point x_0 is an expression of the form

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n = \overset{n=0}{\underset{\uparrow}{a_0}} + \overset{n=1}{a_1(x - x_0)} + \overset{n=2}{a_2(x - x_0)^2} + \dots \quad \text{power series}$$

(Here a_n denotes the coefficient.

$$\rightarrow a_0 + \sum_{n=1}^{\infty} a_n (x - x_0)^n$$

We say that a power series **converges** at the point $x = c$ if

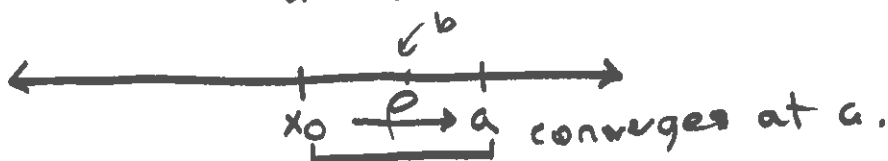
$$\lim_{N \rightarrow \infty} \underbrace{\sum_{n=0}^N a_n (c - x_0)^n}_{\text{finite series}} = L$$

$\lim_{n \rightarrow \infty} S_n = L$

Partial Sums S_n

exists as a finite number. If this limit does not exist, the power series **diverges** at $x = c$.

$$\sum_{n=0}^{\infty} a_n (a - x_0)^n$$



Theorem. There is a number ρ ($0 \leq \rho \leq \infty$) called the radius of convergence of a power series such that the series converges absolutely for $|x - x_0| < \rho$ and diverges for $|x - x_0| > \rho$. If the series converges for all values of x then $\rho = \infty$.

power series

Theorem. Ratio test: If, for n large, the coefficients of a_n are nonzero and satisfy $\lim_{n \rightarrow \infty} \underbrace{\left| \frac{a_n}{a_{n+1}} \right|}_{\text{ratio}} = L$, ($0 \leq L < \infty$)

then the radius of convergence of the power series $\sum_{n=0}^{\infty} a_n (x - x_0)^n$ is $\rho = L$.

ex. Determine the convergence of $\sum_{n=0}^{\infty} \frac{(-2)^n}{n+1} (x-3)^n$

center: 3

$a_n = \frac{(-2)^n}{n+1}$

$$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{(-2)^n}{n+1}}{\frac{(-2)^{n+1}}{n+2}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-2)^n}{n+1} \cdot \frac{n+2}{(-2)^{n+1}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{n+2}{-2(n+1)} \right| = \lim_{n \rightarrow \infty} \frac{1}{2} \left(\frac{n+2}{n+1} \right) = \frac{1}{2} = \rho$$

↑
radius of convergence

converges on $(3 - \frac{1}{2}, 3 + \frac{1}{2}) = (\frac{5}{2}, \frac{7}{2})$

Do not know the convergence at $\frac{5}{2}, \frac{7}{2}$.

$$x = \frac{5}{2}: \sum_{n=0}^{\infty} \frac{(-2)^n}{n+1} \left(\frac{5}{2} - 3 \right)^n = \sum_{n=0}^{\infty} \frac{(-2)^n}{n+1} \left(-\frac{1}{2} \right)^n = \sum_{n=0}^{\infty} \frac{1}{n+1}$$

harmonic series diverges.

$$x = \frac{7}{2}: \sum_{n=0}^{\infty} \frac{(-2)^n}{n+1} \left(\frac{7}{2} - 3 \right)^n = \sum_{n=0}^{\infty} \frac{(-2)^n}{n+1} \left(\frac{1}{2} \right)^n$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n 2^n}{n+1} \left(\frac{1}{2} \right)^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{n+1}$$

alternating harmonic series converges.

convergence on $(\frac{5}{2}, \frac{7}{2}] \leftarrow \text{interval of convergence}$
IOC

$$n!! \begin{cases} \rightarrow n(n-2) \cdots 2 & \text{even} \\ \rightarrow n(n-2) \cdots 1 & \text{odd} \end{cases}$$

$$n! = n(n-1)!$$

$$n! = n(n-1) \cdots 2 \cdot 1$$

ex. Determine the convergence of $\sum_{n=0}^{\infty} \frac{(n+2)!}{n!} (x+2)^n$ center: -2

$$a_n = \frac{(n+2)!}{n!}$$

$$a_n = \frac{(n+2)!}{n!} = \frac{(n+2)(n+1)n!}{n!} = (n+2)(n+1)$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+2)(n+1)}{(n+3)(n+2)} \right| = \lim_{n \rightarrow \infty} \left| \frac{n^2 + 3n + 2}{n^2 + 5n + 6} \right| = 1$$

convergence on $(-2-1, -2+1) = (-3, -1)$

$$x = -3: \sum_{n=0}^{\infty} (n+2)(n+1)(-3+2)^n = \sum_{n=0}^{\infty} (n+2)(n+1)(-1)^n \not\rightarrow 0.$$

diverges.

$$x = -1: \sum_{n=0}^{\infty} (n+2)(n+1)(-1+2)^n = \sum_{n=0}^{\infty} (n+2)(n+1) \not\rightarrow 0$$

diverges.

I.O.C: $(-3, -1)$.

$$\sum_{n=0}^{\infty} a_n (x-x_0)^n$$

• Shifting the summation index

$$\sum_{n=1}^{\infty} (n+1)x^{n-1} = \sum_{k=0}^{\infty} ((k+1)+1)x^{(k+1)-1} = \sum_{k=0}^{\infty} (k+2)x^k$$

$n \rightarrow k = n-1 \rightarrow n = k+1$
 $n=1: k=0$

$$\sum_{n=1}^{\infty} \frac{n}{n+3} x^{n+3} = \sum_{n=4}^{\infty} \frac{(n-3)}{(n-3)+3} x^{(n-3)+3} = \sum_{n=4}^{\infty} \frac{n-3}{n} x^n$$

ex. Write $\sum_{n=1}^{\infty} 2na_n x^{n-1} + \sum_{n=0}^{\infty} 6a_n x^{n+1}$ as a single series.

1. equate powers

2. equate index

we can increase index by removing terms

$$= \sum_{n=0}^{\infty} 2(n+1)a_{n+1} x^n + \sum_{n=1}^{\infty} 6a_{n-1} x^n$$

$$= (2(0+1)a_{0+1} x^0 + \sum_{n=1}^{\infty} 2(n+1)a_{n+1} x^n) + \sum_{n=1}^{\infty} 6a_{n-1} x^n$$

$$= 2a_1 + \sum_{n=1}^{\infty} (2(n+1)a_{n+1} x^n + 6a_{n-1} x^n)$$

$$= 2a_1 + \sum_{n=1}^{\infty} (2(n+1)a_{n+1} + 6a_{n-1}) x^n$$

$$ax^2 + bx + c = 0 \quad \text{for all values}$$

$$\Rightarrow a, b, c = 0.$$

Theorem. If $\sum_{n=0}^{\infty} a_n(x-x_0)^n = 0$ for all x in some open interval, then each coefficient $a_n = 0$.

ex. Determine a_n if

$$y' + y = 0$$

$$\sum_{n=1}^{\infty} \underbrace{na_n}_{\uparrow} x^{n-1} + \sum_{n=-1}^{\infty} \underbrace{a_{n+1}}_{\uparrow} x^{n+1} = 0.$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+1)a_{n+1}x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} \underline{(n+1)a_{n+1} + a_n} x^n = 0$$

$$\Rightarrow \underline{(n+1)a_{n+1} + a_n = 0 \quad \text{for } n \geq 0.}$$

Solve for highest indexed term

$$\Rightarrow (n+1)a_{n+1} = -a_n$$

$$\Rightarrow \boxed{a_{n+1} = \frac{-a_n}{n+1}, \quad n \geq 0} \quad \text{recurrence relation}$$

let a_0 be a constant. Then $n=0: a_1 = \frac{-a_0}{1} = -a_0$

$$n=0: a_1 = -a_0 \longrightarrow a_1 = -1 \cdot a_0$$

$$n=1: a_2 = \frac{-a_1}{2} = \frac{-(-a_0)}{2} = \frac{a_0}{2} \rightarrow a_2 = \frac{1}{2} \cdot a_0$$

$$n=2: a_3 = \frac{-a_2}{3} = -\frac{1}{3} \cdot \frac{1}{2} a_0 = -\frac{1}{6} a_0$$

$$n=3: a_4 = \frac{-a_3}{4} = -\frac{1}{4} \left(-\frac{1}{6} a_0 \right) = \frac{1}{24} a_0$$

$(-1)^n$	+	even
	-	odd
$(-1)^{n+1}$	+	odd
	-	even

$$n=k: a_{k+1} = \frac{-a_k}{k+1} = (-1)^{k+1} \frac{a_0}{(k+1)!}$$

$k \geq 0$
↑
recurrence relation

So

~~$$a_n = (-1)^{n+1} \frac{a_0}{(n+1)!}$$~~

$n = k+1$

a_0 is $n=0$
 a_1 is $n=1 \rightarrow k=0$
 $k \rightarrow n-1$
 $n \rightarrow k+1$

$$a_n = \frac{(-1)^n}{n!} a_0$$

↑
series

$$n=0, a_0 = a_0$$

$$n=1, -1 \cdot a_0 = a_1$$

$$n=2, \frac{1}{2} a_0 = a_2$$

$$\sum_{n=0}^{\infty} a_n x^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} a_0 x^n = \sum_{n=0}^{\infty} a_0 \frac{1}{n!} (-x)^n$$

$$= a_0 e^{-x}$$

Theorem. If $f(x) = \sum_{n=0}^{\infty} a_n(x-x_0)^n$ has a positive radius of convergence R , then f is differentiable in $|x-x_0| < R$ and

$$f'(x) = \sum_{n=1}^{\infty} n a_n (x-x_0)^{n-1} \quad \text{for } |x-x_0| < R.$$

Furthermore, f is integrable in $|x-x_0| < R$ and

$$\int f(x) dx = C + \sum_{n=0}^{\infty} \frac{a_n}{n+1} (x-x_0)^{n+1} \quad \text{for } |x-x_0| < R.$$

ex. Starting with the geometric series

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1,$$

find a power series for each of the following functions:

1. $\ln(1+x)$ $\frac{d}{dx} (\ln(1+x)) = \frac{1}{1+x} = \frac{1}{1-(-x)}$

$$= \sum_{n=0}^{\infty} (-x)^n = \sum_{n=0}^{\infty} (-1)^n x^n \quad |x| < 1$$

$$\text{So } \ln(1+x) = \int \sum_{n=0}^{\infty} (-1)^n x^n dx = \left(\sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} \right) + C$$

if $x=0$: $\ln(1+0) = \sum_{n=0}^{\infty} (-1)^n \frac{0^{n+1}}{n+1} + C = C \Rightarrow C=1$

$0 = \ln(1)$

$$\boxed{\ln(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}}$$

$$2. \frac{1}{(1+x)^2} \rightarrow \int \frac{1}{(1+x)^2} dx = \frac{-1}{1+x} + C$$

$$\Rightarrow \frac{-1}{1+x} = -\frac{1}{1+x} = -\sum_{n=0}^{\infty} (-x)^n = -\sum_{n=0}^{\infty} (-1)^n x^n$$

$$= \sum_{n=0}^{\infty} (-1)^{n+1} x^n$$

$$\text{So } \frac{1}{(1+x)^2} = \frac{d}{dx} \left(\sum_{n=0}^{\infty} (-1)^{n+1} x^n \right) = \sum_{n=1}^{\infty} (-1)^{n+1} n x^{n-1}$$

$$3. \tan^{-1} x \rightarrow \frac{d}{dx} \tan^{-1}(x) = \frac{1}{1+x^2} = \frac{1}{1-(-x^2)}$$

$$= \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

$$\text{So } \tan^{-1}(x) = \int \sum_{n=0}^{\infty} (-1)^n x^{2n} dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} + C$$

$$\tan^{-1}(0) = 0 \Rightarrow C = 0.$$

$$\text{So } \tan^{-1}(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

ex. Find $f'(x)$ for

$$1. f(x) = \frac{1}{(1+x)^2} \rightarrow \frac{d}{dx} \left(\frac{1}{(1+x)^2} \right)$$

$$= \frac{d}{dx} \left(\sum_{n=1}^{\infty} (-1)^{n+1} n x^{n-1} \right)$$

$$= \sum_{n=2}^{\infty} (-1)^{n+1} n(n-1) x^{n-2}$$

represents $\frac{-2}{(1+x)^3}$

$$2. f(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \neq \sinh(x)$$

$$\begin{aligned} f' &= \frac{d}{dx} \left(\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \right) = \sum_{n=0}^{\infty} (-1)^n \frac{(2n+1) x^{2n}}{(2n+1)!} \\ &= \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \quad \rightarrow \frac{2n+1}{(2n+1)(2n)!} x^{2n} \end{aligned}$$

$$= \cosh(x).$$

$$(a+b)(c+d)$$

+

$$\frac{(a+b)(a_1+a_2+a_3+\dots)}{\sin(x)\cos(x)}$$

ex. Find the first three nonzero terms in the power series expansion for the product $f(x)g(x)$ where

$$f(x) = \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}, \quad \text{we want up to } x^5.$$

$$g(x) = \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$$

$$\sin(x)\cos(x) = \left(\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \right) \left(\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \right)$$

$$= \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right) \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right)$$

x^0	x^1	x^2	x^3	x^4	x^5
x			$-\frac{x^3}{2!}$		$\frac{x^5}{4!}$
	x		$-\frac{x^3}{3!}$		$\frac{x^5}{3!2!}$
					$\frac{x^5}{5!}$
$= x + \left(-\frac{1}{2!} - \frac{1}{3!} \right) x^3 + \left(\frac{1}{4!} + \frac{1}{3!2!} + \frac{1}{5!} \right) x^5 + \dots$					
$= x - \frac{2}{3} x^3 + \frac{11}{120} x^5 + \dots$					

Def. A function f is analytic at x_0 if it can be represented by a power series $f(x) = \sum_{n=0}^{\infty} a_n(x-x_0)^n$ with a positive (or an infinite) radius of convergence.

NOTE:

1. In Calculus 2, we have that the Taylor series of f at x_0 is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n.$$

Therefore, if f is analytic at x_0 , then

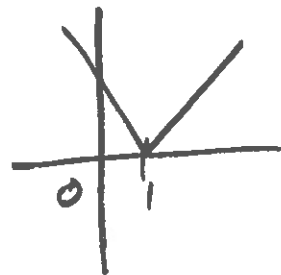
$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n.$$

$x^{7/3}$
↑

2. If a function does not have derivatives of all orders at x_0 , then it cannot be analytic at x_0 .

ex.

1. $f(x) = |x-1|$ is not analytic at $x=1$.



$f'(1)$ DNE

2. $g(x) = x^{7/3}$ is not analytic at $x=0$.

$$g' = \frac{7}{3} x^{4/3}, \quad g'' = \frac{7}{3} \cdot \frac{4}{3} x^{1/3}, \quad g''' = \frac{7}{4} \cdot \frac{4}{3} \cdot \frac{1}{3} x^{-2/3}$$

$g'''(0)$ DNE

<u>$f(x)$</u>	<u>Maclaurin Series</u>	<u>I.O.C.</u>
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$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad (-1, 1)$$

$$\ln(1-x) = -\sum_{n=1}^{\infty} \frac{x^n}{n}, \quad [-1, 1)$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}, \quad (-1, 1]$$

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)}, \quad [-1, 1]$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad (-\infty, \infty)$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \quad (-\infty, \infty)$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad (-\infty, \infty)$$

,